

CATALIN WAACK

AI PRODUCT AND FULL STACK ENGINEER

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PROFILE

I am a product engineer and former CTO who enjoys turning ambitious ideas into reliable products. My work combines native iOS and SwiftUI, full stack systems and hands on AI engineering. I have helped ship a mobile product used by more than 12,000 people and built AI workflows around recovery, observability and human review. I work closely with Product, Design and Backend teams from the first prototype through architecture, testing and live operations.

TECHNICAL SKILLS

Mobile and Apple: Swift, SwiftUI, Swift Testing, XCTest, CoreMotion, Accelerate and vDSP, LiveKit, WebRTC, async and await, Task, TaskGroup, MainActor isolation, state driven UI, UIKit bridges and Swift Package Manager integration.

AI engineering: Claude Code, OpenAI Codex, Gemini CLI, Grok Build, MCP, agent orchestration, prompt engineering, tool policies, retrieval, evaluation, human approval and AI assisted planning, implementation, testing and review.

Architecture and delivery: Modular boundaries, REST and JSON contracts, API compatibility and versioning, Python, FastAPI, TypeScript, PostgreSQL, CI and CD, GitHub Actions, Sentry, Docker and Kubernetes.

PROFESSIONAL EXPERIENCE

BreakEats LLC | Product Engineer, Contract | 2024 to 2026

- Worked directly with a former NVIDIA manager and multiple patent holder on novel audio motion research across iOS, Python and server systems, with SwiftUI interfaces and real time capture from iPhone and headphone sensors.
- Built Swift and CoreMotion capture plus latency aware matching across nine frequency bands using Accelerate and vDSP; connected LiveKit and WebRTC flows with evolving JSON APIs and Python and TypeScript services.
- Used async and await, Task, TaskGroup and MainActor isolation, with Swift Testing and XCTest coverage plus diagnostic timelines that made algorithm errors traceable and tunable.

TailRemote | SwiftUI and XCTest, Independent project | Open source

- Built a SwiftUI iPhone client for direct macOS Screen Sharing over Tailscale, with a stateful gesture canvas, UIKit keyboard bridge and MainActor isolated VNC session state.
- Integrated RoyalVNCKit through Swift Package Manager and added XCTest geometry coverage for gesture and viewport behavior. github.com/C4T4/TailRemote

DAYONE GmbH | Senior Frontend Developer, Contract | 2024

- Developed the FAMFAM React Native mobile app and React dashboard in a cross functional team of more than ten; the product reached more than 12,000 users and the App Store Top 10 in Music.
- Built artist publishing, events, guest lists, invitations, reservations, push notifications and Stripe donations across mobile and web applications backed by AWS APIs.

PROFESSIONAL EXPERIENCE CONTINUED

Navigator Insurance Brokers Ltd | AI Product Consultant and Lead Full Stack Engineer, Contract | 2026

- Owned product discovery, prototyping, technical architecture and rollout for ElectaCar, turning an early electric vehicle platform concept into a reliable live product.
- Built a Python and FastAPI backend combining government pricing, technical specifications, insurer mappings and media into one catalog covering more than 100 vehicle models.
- Designed a 12 stage AI research and publishing workflow with source validation, evaluation, retries, restart recovery, observability and human approval, separated across public API, AI and browser worker boundaries.

HeyAmigo and AmigoEngine, Independent Products | Creator and AI Product Engineer | 2026

- Built and published a multi provider agent runtime that executes long lived Claude Code, OpenAI Codex, Gemini CLI and Grok Build tasks from WhatsApp and Telegram; the package has more than 15,000 npm downloads.
- Defined system prompts, memory rules, tool policies and provider conventions as version controlled engineering artifacts, with prompt logging, session recovery and isolated MCP browser workers.
- Designed durable queues, stable task identity, idempotent side effects, worker heartbeats and restart recovery so concurrent agent work remains ordered, observable and safe to resume.

Syntax GmbH für Aus- und Weiterbildung | Lead Product Engineer, Contract | 2023 to 2024

- Built and launched a consumer learning platform from concept to paying customers in three months for a provider with more than 100 instructors; it has processed more than 27,000 quiz results.
- Owned product discovery, architecture, API integration, testing and rollout for course access, progress, assessments, certificates and Stripe purchases across learner and operations workflows.

VoxR GmbH | CTO, Employee with Revenue Share | 2017 to 2020

- Led product and engineering for a live event platform used by the German federal government and 7 of the 10 largest DAX companies, including events with more than 10,000 participants.
- Designed for burst traffic and unreliable venue connectivity while collaborating directly with customers, product stakeholders and delivery teams.

EDUCATION

Computer Science studies | Technische Universität Braunschweig | 2014 to 2016

Government certified technical assistant for IT | BSZ Stendal | 2011 to 2013

LANGUAGES

German: native | English: professional | Spanish: intermediate